

Lipschitz Periodic Shadowing

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Shadowing and Lipschitz Shadowing

- $f : M \mapsto M$, $f \in C^1$, $M \in C^\infty$, dist
- $\xi = \{x_n\}$ is a d -pseudotrajectory, if

$$\text{dist}(x_{n+1}, f(x_n)) < d.$$

- Shadowing property (POTP)
 $\forall \epsilon > 0 \exists d > 0$ such that $\forall d$ -pseudotrajectory ξ there exists an exact trajectory $\{p_n\}$ such that

$$\text{dist}(x_n, p_n) < \epsilon.$$

- Lipschitz shadowing property (LipSP)
 $\exists L, d_0 > 0$ such that $\forall d < d_0$ and d -pseudotrajectory ξ there exists an exact trajectory $\{p_n\}$ such that

$$\text{dist}(x_n, p_n) < Ld.$$

Periodic shadowing and structural stability

- Periodic shadowing (PerSh)

$\forall \epsilon > 0 \exists d > 0$ such that $\forall$ periodic d -pseudotrajectory ξ there exists a periodic exact trajectory $\{p_n\}$ such that

$$\text{dist}(x_n, p_n) < \epsilon.$$

- Lipschitz periodic shadowing (LipPerSh)

PerSh with $\epsilon = Ld$

- S — the set of structurally stable diffeomorphisms

There exists a neighborhood U of f in the C^1 -topology such that $\forall g \in U$ the diffeomorphisms f and g are topologically conjugate.

- ΩS — the set of Ω -stable diffeomorphisms

There exists a neighborhood U of f in the C^1 -topology such that $\forall g \in U$ the diffeomorphisms f and g are topologically conjugate on their nonwandering sets.

Known Facts and Results

- $\text{Int}^1(\text{POTP}) = S$ (Sakai, 1994)
- $\text{POTP} \neq S$
- $\text{PerSh} \neq \Omega S$
- $\text{LipSP} = S$ (Pilyugin, Tikhomirov, 2009)
- variational shadowing is equivalent to structural stability (Pilyugin, 2009)

Main results

Theorems (Osipov, Pilyugin, Tikhomirov, 2009)

- $\text{Int}^1(\text{PerSh}) = \Omega S$
- $\text{LipPerSh} = \Omega S$

General Scheme of the Proof

- $\Omega S \subset \text{LipPerSh}$
- $\text{Int}^1(\text{PerSh}) \subset \Omega S$
- $f \in \text{LipPerSh} \Rightarrow f \in \Omega S$
 - Step 1. hyperbolicity of periodic points
 - Step 2. uniform hyperbolicity of periodic points
 - Step 3. f has the Axiom A
 - Step 4. f satisfies the no-cycle condition

Proof of $\Omega S \subset \text{LipPerSh}$

- Spectral decomposition theorem:
 $\Omega(f) = \Omega_1 \cup \dots \cup \Omega_m$, Ω_j is hyperbolic and has a dense semi-trajectory
- ξ is a periodic d -pseudotrajectory, $\xi \subset U(\Omega_j)$ for some j
- Shadowing lemma:
if Λ is hyperbolic then f has LipSh and is expansive in some $U(\Lambda)$

Proof of $\text{Int}^1(\text{PerSh}) \subset \Omega S$

- HP — set of diffeomorphisms f such that every periodic point of f is hyperbolic
Lemma (Aoki, 1992, Hayashi, 1992). $\text{Int}^1(\text{HP}) = \Omega S$
- It is enough to prove that $\text{Int}^1(\text{PerSh}) \subset \text{HP}$
- h is a C^1 -small perturbation of f that is linear in $U(p)$, p is a nonhyperbolic periodic point for h

Proof of $\text{LipPerSh} \subset \Omega S$, Steps 1 and 2

- $f, f^{-1} \in \text{LipPerSh}$ with $L > 1$
- Lemma: Every periodic point is hyperbolic
- Key lemma: Set of all periodic points of f has all properties of a standard hyperbolic set except compactness.

$$|Df^j(p)v_s| \leq C\lambda^j|v_s|, \quad |Df^{-j}(p)v_u| \leq C\lambda^j|v_u|,$$

where $j \geq 0$, $v_s \in S(p)$, $v_u \in U(p)$

- p is an m -periodic point, let $v_0 = v_u \in U(p)$, $v_{i+1} = Df^i(p)v_i$,

$$\lambda_i = |v_{i+1}|/|v_i|, \quad a_0 = \tau, \quad a_{i+1} = \lambda_i a_i - 1,$$

where τ is chosen such that $a_m = 0$

- $w_i = a_i v_i / |v_i|$ for $0 \leq i \leq m-1$, $\{w_i\}$ is an $m(n+1)$ -periodic
- $|Df^i(p)v_u| = \lambda_0 \cdots \lambda_{i-1} > \frac{1}{16L} \left(1 + \frac{1}{8L}\right)^i |v_u|$, $0 \leq i \leq m-1$.

Proof of $\text{LipPerSh} \subset \Omega S$, Steps 3 and 4

- Lemma: f satisfies the Axiom A
 - P_I — the set of periodic points of index I
 - ClP_I is a hyperbolic set.
 - density of periodic points in $\Omega(f)$
- Lemma: f has no cycles
 - any cycle is approximated by periodic pseudotrajectories
 - any cycle is approximated by periodic exact trajectories

Conclusion

Theorems (Osipov, Pilyugin, Tikhomirov, 2009)

- $\text{Int}^1(\text{PerSh}) = \Omega S$
- $\text{LipPerSh} = \Omega S$

Thank you very much for your attention!